



Newbear 77-68

DEMON Monitor



DEMON is a 2KB monitor program for the Newbear 77-68.

Writing DEMON was first started in 1979, by Chris Carter using a cross-compiler written by Mike Davies on a PDP 11/34 for use on Newbear 77/68s built by Mike and Chris's brother Dave. DEMON got put on hold for nearly 40 years and the SWI support was finally finished in 2018 !

The main aims for this program were:-

1) Provide most of the features of the **MINIMON** monitor along with the ability to read and write **.S19** files of the **MIKBUG** monitor along with some other features.

Commands implemented in V5 of DEMON are:-

B = Block Move	Based on code from Minimon
M = Modify	" " " " Minimon
D = Dump	" " " " Minimon
L = S19 format file load	" " " " Mikbug
P = File : `punch` S19 format	" " " " Mikbug
G = Go <addr>	
? = Show version and Help text	
I = On/Off toggle for LED count during input wait	
Z = Set memory to <switches>	/__ For debugging
Y = Check memory for <switches>	\ corrupted memory
X = Find byte on <switches>	
T = Test memory (R/W 0 to 255)	
S = <switches> to LEDs	

2) Rather than an application having to JMP or JSR to support routines in the monitor for I/O etc, DEMON aimed to use **SWI** calls (**So**ft**W**are **I**nterrupts) for these calls, so that application programs did not need to know what actual memory address these routines were, meaning that changes to the monitor did not affect application code.

What an application has to do is to issue an **SWI** instruction, **followed by a code byte** indicating what function is to be performed. DEMON performs the work, then returns to the instruction following the function byte.

Eg:- SWI ; Call DEMON to....
 \$01 ; read character into A

Main SWI call functions are:-

\$00 = Back to DEMON, reset stack
\$01 = Get char into A from ACIA
\$03 = Get char into A from ACIA and echo it
\$04 = Output character from A
\$05 = Output string pointed to by X (End-string = any byte < \$07)
\$06 = Output string immediately following this \$06

For other functions, see DEMON listing